Compact Equations of Motion

Equation	Compact Form
Cabin Bounce (z_c)	$m_c\ddot{z}_c + \sum C_c(\dot{z}_c - \dot{z}_s) + \sum K_c(z_c - z_s)$ $+ \text{Pitch/Roll Terms} = 0$
Cabin Pitch (θ_c)	$I_{yyc}\ddot{\theta}_c + \text{Damping Coupling} + \text{Stiffness Coupling}$ $- M_c g h_{cp}\theta_c = 0$
Cabin Roll (ϕ_c)	$I_{xxc}\ddot{\phi}_c + \text{Bounce/Pitch Coupling} + \text{Roll Terms} = 0$
Sprung Mass (z_s)	$M_s\ddot{z}_s + K_f(z_s - a\theta_s - z_{1f}) + C_f(\dot{z}_s - a\dot{\theta}_s - \dot{z}_{1f}) + \sum_{i=2}^{3}(K_i + C_i) = 0$
Sprung Mass (θ_s)	$I_{syy}\ddot{\theta}_s + I_{syx}\ddot{\phi}_s - a(F_f) + \sum_{i=2}^{3} l_i(F_i) = 0$
Sprung Mass (ϕ_s)	$-(I_{sxx} + m_s h_1^2)\ddot{\phi}_s + I_{sxy}\ddot{\theta}_s = -m_s g h_1 \phi_s$ $+ \text{Roll Suspension Terms}$
Geometric Relations	$(z_s + l_i\theta + S_{tfi}\phi) - (L_D + \Delta l)\sin(\beta_i - \theta) = (z_{us})_i$

Table 1: Summary of Compact Equations of Motion

Final Set of Equations

1. Cabin Bounce (z_c)

$$
\begin{aligned}
m_c\ddot{z}_c &+ (C_{cfl} + C_{cfr} + C_{crl} + C_{crr})\dot{z}_c \\
&- (C_{cfl} + C_{cfr} + C_{crl} + C_{crr})\dot{z}_s \\
&+ (K_{cfl} + K_{cfr} + K_{crl} + K_{crr})z_c \\
&- (K_{cfl} + K_{cfr} + K_{crl} + K_{crr})z_s \\
&- (C_{cfl}l_{cfcg} + C_{cfr}l_{cfcg} - C_{crl}l_{crcg} + C_{crr}l_{crcg})\dot{\theta}_c \\
&- (-C_{cfl}l_{cf} + C_{cfr}l_{cf} - C_{crl}l_{cr} - C_{crr}l_{cr})\dot{\theta}_s \\
&- (-C_{cfl}b - C_{cfr}a - C_{crl}b - C_{crr}a)\dot{\phi}_c \\
&- (C_{cfl}b - C_{cfr}a + C_{crl}b - C_{crr}a)\dot{\phi}_s \\
&- (K_{cfl}l_{cfcg} + K_{cfr}l_{cfcg} - K_{crl}l_{crcg} - K_{crr}l_{crcg})\theta_c \\
&- (-K_{cfl}l_{cf} - K_{cfr}l_{cf} - K_{crl}l_{cr} - K_{crr}l_{cr})\theta_s \\
&- (-K_{cfl}b + K_{cfr}a - K_{crl}b + K_{crr}a)\phi_c \\
&- (K_{cfl}b - K_{cfr}a + K_{crl}b - K_{crr}a)\phi_s = 0
\end{aligned} \tag{1}
$$

Cabin Pitch (θ_c)

$$
\begin{aligned}
I_{yyc}\ddot{\theta}_c &- (C_{cfl}l_{cfcg} + C_{cfr}l_{cfcg} - C_{crl}l_{crcg} - C_{crr}l_{crcg})\dot{z}_c \\
&- (-C_{cfl}l_{cfcg} - C_{cfr}l_{cfcg} - C_{crl}l_{crcg} - C_{crr}l_{crcg})\dot{z}_s \\
&- (-C_{cfl}l_{cfcg}^2 - C_{cfr}l_{cfcg}^2 - C_{crl}l_{crcg}^2 - C_{crr}l_{crcg}^2)\dot{\theta}_c \\
&- (C_{cfl}l_{cfcg}l_{cf} + C_{cfr}l_{cfcg}l_{cf} - C_{crl}l_{crcg}l_{cr} - C_{crr}l_{crcg}l_{cr})\dot{\theta}_s \\
&- (-C_{cfl}l_{cfcg}b + C_{cfr}l_{cfcg}a - C_{crl}l_{crcg}b + C_{crr}l_{crcg}a)\dot{\phi}_c \\
&- (C_{cfl}l_{cfcg}b - C_{cfr}l_{cfcg}a + C_{crl}l_{crcg}b - C_{crr}l_{crcg}a)\dot{\phi}_s \\
&- (K_{cfl}l_{cfcg} + K_{cfr}l_{cfcg} - K_{crl}l_{crcg} - K_{crr}l_{crcg})z_c \\
&- (-K_{cfl}l_{cfcg} - K_{cfr}l_{cfcg} - K_{crl}l_{crcg} - K_{crr}l_{crcg})z_s \\
&- (-K_{cfl}l_{cfcg}^2 - K_{cfr}l_{cfcg}^2 - K_{crl}l_{crcg}^2 - K_{crr}l_{crcg}^2 + M_cgh_{cp})\theta_c \\
&- (K_{cfl}l_{cfcg}l_{cf} + K_{cfr}l_{cfcg}l_{cf} - K_{crl}l_{crcg}l_{cr} - k_{crr}l_{crcg}l_{cr})\theta_s \\
&- (-K_{cfl}l_{cfcg}b + K_{cfr}l_{cfcg}a - K_{crl}l_{crcg}b + K_{crr}l_{crcg}a)\phi_c \\
&- (K_{cfl}l_{cfcg}b - K_{cfr}l_{cfcg}a + K_{crl}l_{crcg}b - K_{crr}l_{crcg}a)\phi_s = 0
\end{aligned} \tag{2}
$$

3. Cabin Roll (ϕ_c)

$$
\begin{aligned}
I_{xxc}\ddot{\phi}_c &- (-C_{cfl}b + C_{cfr}a - C_{crl}b + C_{crr}a)\dot{z}_c \\
&- (C_{cfl}b - C_{cfr}a + C_{crl}b - C_{crr}a)\dot{z}_s \\
&- (-C_{cfl}l_{cfcg}b - C_{cfr}l_{cfcg}a + C_{crl}l_{crcg}b + C_{crr}l_{crcg}a)\dot{\theta}_c \\
&- (C_{cfl}l_{cfcg}b + C_{cfr}l_{cfcg}a - C_{crl}l_{crcg}b - C_{crr}l_{crcg}a)\dot{\theta}_s \\
&- (-C_{cfl}b^2 + C_{cfr}a^2 - C_{crl}b^2 + C_{crr}a^2)\dot{\phi}_c \\
&- (C_{cfl}b^2 - C_{cfr}a^2 + C_{crl}b^2 - C_{crr}a^2)\dot{\phi}_s \\
&- (-K_{cfl}b + K_{cfr}a - K_{crl}b + K_{crr}a)z_c \\
&- (K_{cfl}b - K_{cfr}a + K_{crl}b - K_{crr}a)z_s \\
&- (-K_{cfl}l_{cfcg}b - K_{cfr}l_{cfcg}a + K_{crl}l_{crcg}b + K_{crr}l_{crcg}a)\theta_c \\
&- (K_{cfl}l_{cfcg}b + K_{cfr}l_{cfcg}a - K_{crl}l_{crcg}b - K_{crr}l_{crcg}a)\theta_s \\
&- (-K_{cfl}b^2 + K_{cfr}a^2 - K_{crl}b^2 + K_{crr}a^2)\phi_c \\
&- (K_{cfl}b^2 - K_{cfr}a^2 + K_{crl}b^2 - K_{crr}a^2)\phi_s = 0
\end{aligned} \tag{3}
$$

4. Sprung Mass Bounce

$$
\begin{aligned}
M_s\ddot{z}_s &+ K_f(z_s - a\theta_s - z_{1f}) + C_f(\dot{z}_s - a\dot{\theta}_s - \dot{z}_{1f}) \\
&+ K_2(z_s - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta_s) \\
&+ C_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}_s) \\
&+ K_3(z_s - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta_s) \\
&+ C_3(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}_s) = 0
\end{aligned} \tag{4}
$$

5. Sprung Mass Pitch

$$
\begin{aligned}
I_{syy}\ddot{\theta}_s + I_{syx}\ddot{\phi}_s &- a[K_f(z_s - a\theta_s - z_{1f}) + C_f(\dot{z}_s - a\dot{\theta}_s - \dot{z}_{1f})] \\
&+ l_2[K_2(z_s - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta_s) \\
&+ C_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}_s)] \\
&+ l_3[K_3(z_s - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta_s) \\
&+ C_3(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}_s)] = 0
\end{aligned} \tag{5}
$$

6. Sprung Mass Roll

$$
\begin{aligned}
-(I_{sxx}\ddot{\phi}_s + m_s h_1^2\ddot{\phi}_s) + I_{sxy}\ddot{\theta}_s = &-m_s g h_1\phi_s \\
&+ k_{tf}(\phi_s - \phi_f) + c_{tf}(\dot{\phi}_s - \dot{\phi}_f) \\
&+ k_{r1}(\phi_s - \phi_{r1} - \phi_{NRS}) + c_{r1}(\dot{\phi}_s - \dot{\phi}_{r1}) \\
&+ k_{r2}(\phi_s - \phi_{r2} - \phi_{NRS}) + c_{r2}(\dot{\phi}_s - \dot{\phi}_{r2})
\end{aligned} \tag{6}
$$

where, for i = 2,3

$$\phi_{NRS} = (\beta_{Li}L_{DLi} - \beta_{Ri}L_{DRi})/S_{tfi} \tag{7}$$

Geometric Relations

$$(z_s + l_{i2}\theta + S_{tf2}\phi) - (L_D + \Delta l)\sin(\beta_{2L} - \theta) = (z_{us})_{2L} \tag{8}$$

$$(z_s + l_{i3}\theta + S_{tf3}\phi) - (L_D + \Delta l)\sin(\beta_{3L} - \theta) = (z_{us})_{3L} \tag{9}$$

4. Sprung Mass Bounce

$$
\begin{aligned}
M_s\ddot{z}_s &- (C_{cfl} + C_{cfr} + C_{crl} + C_{crr})\dot{z}_c \\
&- (-C_{cfl}l_{cfcg} - C_{cfr}l_{cfcg} + C_{crl}l_{crcg} + C_{crr}l_{crcg})\dot{\theta}_c \\
&- (-C_{cfl} - C_{cfr} - C_{crl} - C_{crr})\dot{z}_s \\
&- (C_{cfl}l_{cf} + C_{cfr}l_{cf} + C_{crl}l_{cr} + C_{crr}l_{cr})\dot{\theta}_s \\
&- (K_{cfl} + K_{cfr} + K_{crl} + K_{crr})z_c \\
&- (-K_{cfl}l_{cfcg} - K_{cfr}l_{cfcg} + K_{crl}l_{crcg} + K_{crr}l_{crcg})\theta_c \\
&- (-K_{cfl} - K_{cfr} - K_{crl} - K_{crr})z_s \\
&- (K_{cfl}l_{cf} + K_{cfr}l_{cf} + K_{crl}l_{cr} + K_{crr}l_{cr})\theta_s \\
&+ K_f(z_s - l_f\theta_s - z_{1f}) + F_{df} \\
&+ K_2(z_s - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta_s) \\
&+ C_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}_s) \\
&+ K_3(z_s - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta_s) \\
&+ C_3(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}_s) = 0
\end{aligned}
\tag{10}
$$

5. Sprung Mass Pitch

$$
\begin{aligned}
I_{syy}\ddot{\theta}_s + I_{syx}\ddot{\phi}_s &- (C_{cfl}l_{cfcg} + C_{cfr}l_{cfcg} - C_{crl}l_{crcg} - C_{crr}l_{crcg})\dot{z}_c \\
&- (-C_{cfl}l_{cfcg}^2 - C_{cfr}l_{cfcg}^2 - C_{crl}l_{crcg}^2 - C_{crr}l_{crcg}^2)\dot{\theta}_c \\
&- (-C_{cfl}l_{cf} - C_{cfr}l_{cf} - C_{crl}l_{cr} - C_{crr}l_{cr})\dot{z}_s \\
&- (C_{cfl}l_{cfcg}l_{cf} + C_{cfr}l_{cfcg}l_{cf} - C_{crl}l_{crcg}l_{cr} - C_{crr}l_{crcg}l_{cr})\dot{\theta}_s \\
&- (K_{cfl}l_{cf} + K_{cfr}l_{cf} + K_{crl}l_{cr} + K_{crr}l_{cr})z_c \\
&- (-K_{cfl}l_{cfcg}l_{cf} - K_{cfr}l_{cfcg}l_{cf} + K_{crl}l_{crcg}l_{cr} + K_{crr}l_{crcg}l_{cr})\theta_c \\
&- (-K_{cfl}l_{cf} - K_{cfr}l_{cf} - K_{crl}l_{cr} - K_{crr}l_{cr})z_s \\
&- (K_{cfl}l_{cf}^2 + K_{cfr}l_{cf}^2 + K_{crl}l_{cr}^2 + K_{crr}l_{cr}^2)\theta_s \\
&- l_f\left[K_f(z_s - l_f\theta_s - z_{1f}) + F_{df}\right] \\
&+ l_2\Big[K_2(z_s - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta_s) \\
&\qquad + C_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}_s)\Big] \\
&+ l_3\Big[K_3(z_s - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta_s) \\
&\qquad + C_3(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}_s)\Big] = 0
\end{aligned}
\tag{11}
$$

Setting : Laden	**Freq bounds**	**Min**	**Max**
KF	1 – 2 Hz	210995.44	843981.76
CF	0.1 – 0.4 (cc)	10074.27407	40297.09627
K2	1 – 4.5 Hz	319253.848	6464890.422
K3	2 – 4.5 Hz	357114.712	7231572.918
Csy		0.2	0.4
assym ratio		2.3	4
gamma_c		0.08	0.16
gamma_r		0.08	0.1

Comparison	**Multi Pareto 1 (Best_total)**	**Multi Pareto 2 (Best_longitudinal)**	**Single Opt**
rms_x	0.0447	0.0436	0.0832
rms_y	0.0299	0.0267	0.0414
rms_z	0.7251	0.7364	1.0106
combined	0.7271	0.7381	1.0148

Parameter	**Pareto 1**	**Pareto 2**	**Single**	Δ(P1 - S)	Δ(P2 - S)	Δ(P1 - P2)
K_f	830241	833347	827321	2920	6026	-3106
C_f	31696.6	31696.1	25660	6036.6	6036.1	0.45
K_2	1.14E+06	7.63E+05	2.04E+06	-892,990	-1,272,643	379,656
K_3	650150	1,050,962	862,484	-212,334	188,478	-400,812
cs_minus	0.34714	0.34655	0.34546	0.00168	0.00109	0.0006
asym_ratio	2.4908	2.4917	3.6577	-1.1669	-1.1660	-0.00085
gamma_c	0.13148	0.13048	0.10278	0.0287	0.0277	0.001
gamma_r	0.093	0.0926	0.09593	-0.00293	-0.00333	0.0004

Setting : Laden	**Freq bounds**	**Min**	**Max**
KF	1.4 – 1.6 Hz	416,871.90	534,013.38
CF	0.2 – 0.4 (cc)	6,600.00	21,000.00
K2	1.6 – 1.8 Hz	961,237.04	1,202,623.92
K3	1.6 – 1.8 Hz	961,237.04	1,202,623.92
C-		0.2	0.4
assym ratio		2.3	4
gamma_c		0.08	0.16
gamma_r		0.08	0.1

Comparison	**Option 1**	**Option 2**	**Δ (Opt2 - Opt1)**
rms_x	0.0864	0.0842	-0.0022
rms_y	0.0526	0.0531	0.0005
rms_z	1.0584	1.0651	0.0067
combined	1.0632	1.0698	0.0066

Parameter	**Option 1**	**Option 2**	**Δ (Opt2 - Opt1)**
K_f	481,103.56	463,823.52	-17,280.04
C_f	20,644.52	20,719.31	74.79
K_2	1,138,522.19	995,738.29	-142,783.90
K_3	1,020,694.44	1,022,640.96	1,946.52
cs_minus	0.3305	0.4	0.0695
asym_ratio	3.8753	3.6898	-0.1855
gamma_c	0.1043	0.16	0.0557
gamma_r	0.085	0.1	0.015

$$L = \lambda_s L_{\text{state}} + \lambda_a L_{\text{accel}} + \lambda_r L_{\text{rms}} + \lambda_p L_{\text{phys}} + 2L_{ac}$$